



BIKESTORES

SQL BUSINESS ANALYSIS

SQL MILESTONE PROJECT

EXPLORING DATA. EXTRACTING INSIGHTS. DRIVING DECISIONS.



STORE
PERFORMANCE



CUSTOMER
INSIGHTS



PRODUCT
ANALYSIS



STAFF
PERFORMANCE



BUSINESS
GROWTH

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BUSINESS OBJECTIVES

Key goals of this SQL analysis project



01

ANALYZE STORE PERFORMANCE

Evaluate store-wise sales and product distribution to identify top performing stores.



02

UNDERSTAND CUSTOMER BEHAVIOR

Analyze customer spending patterns, order behavior and loyalty trends.



03

EVALUATE PRODUCT PERFORMANCE

Identify best selling, high value and low performing products across categories.



04

ASSESS STAFF PERFORMANCE

Measure staff efficiency by analyzing the number of orders handled and performance levels.



05

DRIVE BUSINESS INSIGHTS

Extract meaningful insights from data to support strategic decision making.



Turning Data into Insights. Driving Better Decisions.

DATABASE ARCHITECTURE

BikeStores – Relational Database Schema Overview

KEY HIGHLIGHTS



Normalized
Relational
Database



Primary Keys
Ensure Uniqueness



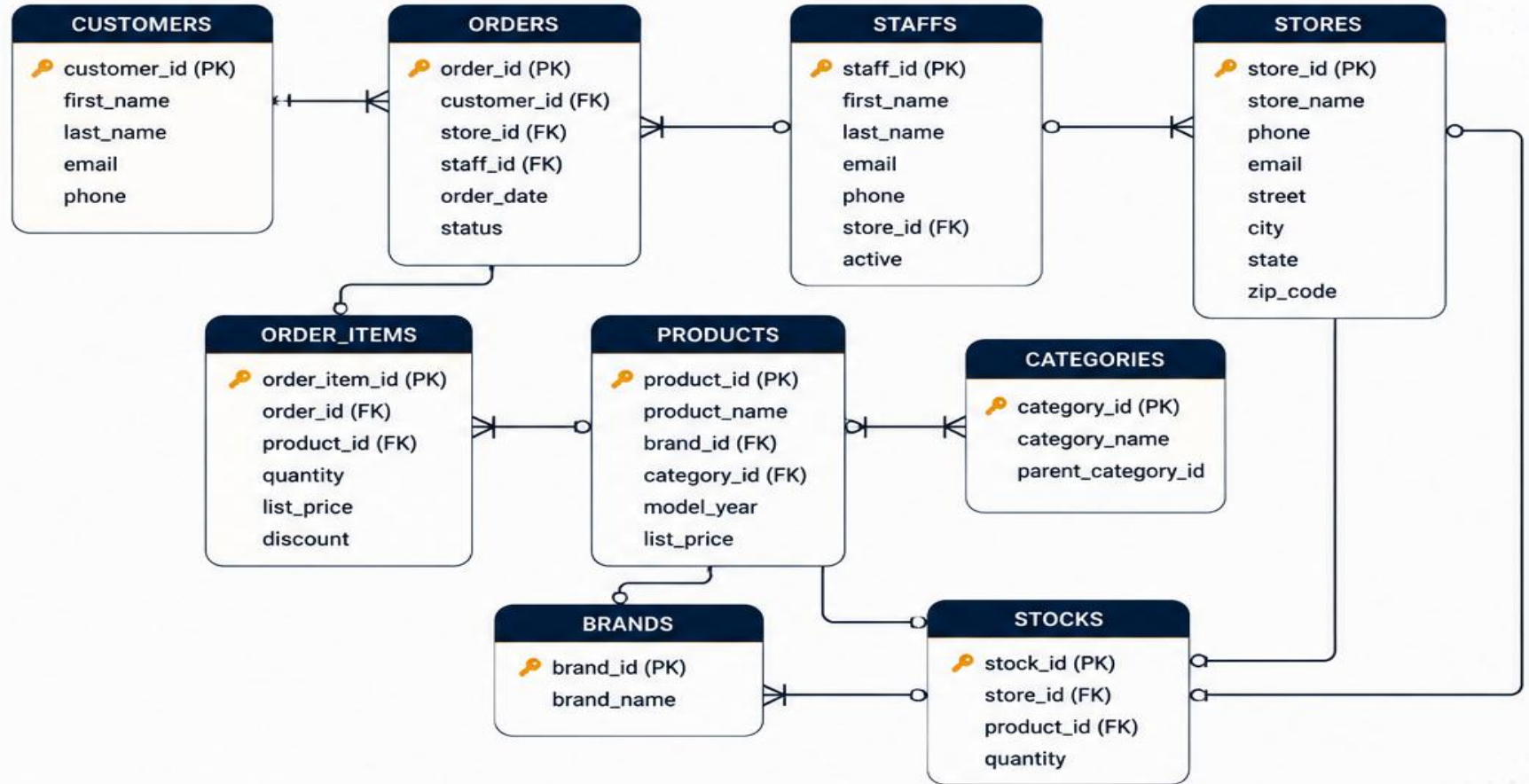
Foreign Keys
Maintain Referential
Integrity



Designed for
Sales, Inventory
and Operations
Analysis

RELATIONSHIP LEGEND

- One to One
- One to Many
- Many to Many



9

TABLES



9

PRIMARY KEYS



11

FOREIGN KEYS



CORE

BUSINESS ENTITIES



100%

RELATIONAL
INTEGRITY



SQL TECHNIQUES USED

Core SQL Concepts Applied in This Project

01



JOIN

Combine data from multiple tables using various types of JOINS to retrieve meaningful results.

02



GROUP BY

Group rows that have the same values into summary rows to perform aggregate calculations.

03



HAVING

Filter grouped data based on aggregate conditions to refine the result set.

04



SUBQUERY

Use nested queries inside other queries to perform complex and powerful analysis.

05



WINDOW FUNCTION

Perform advanced calculations across a set of table rows related to the current row.

06



EXISTS / NOT EXISTS

Check for the existence or absence of rows in a subquery resulting in efficient filtering.

07



AGGREGATE FUNCTIONS

Use functions like SUM, COUNT, AVG, MAX, MIN etc. to calculate summary information.

08



ORDER BY & LIMIT

Sort the result set and limit the number of rows to extract top or specific records.



These SQL techniques help extract valuable insights and solve real business problems efficiently from the BikeStores database.

QUESTION 1

TOTAL PRODUCTS SOLD BY EACH STORE



QUESTION

Find the total number of products sold by each store along with the store name.



SQL QUERY

```
SELECT
    s.store_name, SUM(oi.quantity) AS Total_Products_Sold
FROM
    stores s
JOIN
    orders o USING (store_id)
JOIN
    order_items oi USING (order_id)
GROUP BY s.store_name
ORDER BY Total_Products_Sold DESC;
```



Tables Used: stores, orders, order_items



RESULT

store_name	Total_Products_Sold
Baldwin Bikes	4779
Santa Cruz Bikes	1516
Rowlett Bikes	783



Total **3 rows** returned.
This means data for **3** stores is fetched.



KEY INSIGHT: Baldwin Bikes has sold the highest number of products, followed by Santa Cruz Bikes and Rowlett Bikes.



QUESTION 2

CUMULATIVE SUM OF QUANTITIES SOLD FOR EACH PRODUCT OVER TIME



QUESTION

Calculate the cumulative sum of quantities sold for each product over time.



SQL QUERY

```
SELECT
  p.product_name,
  o.order_date,
  oi.quantity,
  SUM(oi.quantity) OVER (
    PARTITION BY p.product_id
    ORDER BY o.order_date, o.order_id
  ) AS cumulative_quantity
FROM products p
JOIN order_items oi USING(product_id)
JOIN orders o USING(order_id)
ORDER BY p.product_name, o.order_date;
```



Tables Used: products, order_items, orders



RESULT

product_name	order_date	quantity	cumulative_quantity
Electra Amsterdam Fashion 3i Ladies' - 2017/2018	2018-01-01	1	1
Electra Amsterdam Fashion 3i Ladies' - 2017/2018	2018-01-21	2	3
Electra Amsterdam Fashion 3i Ladies' - 2017/2018	2018-04-30	2	5
Electra Amsterdam Fashion 7i Ladies' - 2017	2017-01-29	2	2
Electra Amsterdam Fashion 7i Ladies' - 2017	2017-02-28	1	3
...	...	...	...
Sun Bicycles Cantare 3	2023-12-30	1	15



Total **4722** rows returned.

This shows the running (cumulative) total quantity sold for each product based on the order date and order sequence.



KEY INSIGHT: This cumulative view helps understand the sales trend of each product over time and how quantities have grown with each order.



QUESTION 3

HIGHEST TOTAL SALES PRODUCT FOR EACH CATEGORY



QUESTION

Find the product with the highest total sales (quantity * price) for each category.



SQL QUERY

```
WITH Product_Sales AS (  
  SELECT  
    c.category_name,  
    p.product_name,  
    SUM(oi.quantity * oi.list_price) AS Total_Sales,  
    RANK() OVER(  
      PARTITION BY c.category_id  
      ORDER BY SUM(oi.quantity * oi.list_price) DESC  
    ) AS rnk  
  FROM categories c  
  JOIN products p USING(category_id)  
  JOIN order_items oi USING(product_id)  
  GROUP BY c.category_id, c.category_name, p.product_id, p.product_name  
)  
SELECT  
  category_name, product_name, Total_Sales  
FROM Product_Sales  
WHERE rnk = 1;
```



Tables Used: categories, products, order_items



RESULT

	category_name	product_name	Total_Sales
	Children Bicycles	Electra Boys' 1 (16-inch)	615963.50
	Cycling Clothing	Forte MT-01 Mountain Bike Socks	411954.83
	Cycling Components	SHIMANO Ultegra CS-6500 9-Speed Cassette	226384.50
	Mountain Bikes	Raleigh Talus 7.2 - 2016	2807824.00
	Road Bikes	Trek Domane SLR 9 - 2018	5413020.53
	Touring Bikes	Surly Long Haul Trucker - 2016	1678509.06
	Miscellaneous	Park Tool 106 Repair Stand	3295624.00



Total 7 rows returned.

This shows the top selling product (by total sales) in each category.



KEY INSIGHT: These are the best performing products in each category based on total sales (quantity * price). This insight helps in inventory planning and promotional focus.





QUESTION 4

CUSTOMER WHO SPENT THE MOST MONEY ON ORDERS



QUESTION

Find the customer who spent the most money on orders.



SQL QUERY

```
SELECT
  CONCAT(c.first_name, ' ', c.last_name) AS Customers_Name,
  SUM(oi.quantity * oi.list_price) AS Total_Money
FROM
  customers c
  JOIN
  orders o USING (customer_id)
  JOIN
  order_items oi USING (order_id)
GROUP BY Customers_Name
ORDER BY Total_Money DESC
LIMIT 1;
```



Tables Used: customers, orders, order_items



RESULT

Customers_Name	Total_Money
Pamelia Newman	3780184.00



Top customer by total amount spent:
Pamelia Newman with **\$3,780,184.00**



KEY INSIGHT: Pamela Newman is the top spender, contributing the highest revenue among all customers.



QUESTION 5

HIGHEST-PRICED PRODUCT FOR EACH CATEGORY



QUESTION

Find the highest-priced product for each category name.



SQL QUERY

```
SELECT
    c.category_name, p.product_name, p.list_price
FROM
    categories c
JOIN
    products p USING (category_id)
WHERE
    p.list_price = (SELECT
                    MAX(p2.list_price)
                    FROM
                        products p2
                    WHERE
                        p2.category_id = p.category_id);
```



Tables Used: categories, products



RESULT

	category_name	product_name	list_price
	Children Bicycles	Electra Straight 8 3 (20-inch) - Boy's - 2017	48999.00
	Children Bicycles	Electra Townie 3i EQ (20-inch) - Boy's - 2017	48999.00
	Children Bicycles	Trek Superfly 24 - 2017/2018	48999.00
	Comfort Bicycles	Electra Townie Go! 8i - 2017/2018	259999.00
	Cruisers Bicycles	Electra Townie Commute Go! - 2018	299999.00
	Cruisers Bicycles	Electra Townie Commute Go! Ladies' - 2018	299999.00
	Cyclocross Bicycles	Trek Boone 7 Disc - 2018	399999.00
	Electric Bikes	Trek Powerfly 8 FS Plus - 2017	499999.00
	Electric Bikes	Trek Powerfly 7 FS - 2018	499999.00
	Electric Bikes	Trek Super Commuter+ 8S - 2018	499999.00
	Mountain Bikes	Trek Fuel EX 9.8 27.5 Plus - 2017	529999.00
	Mountain Bikes	Trek Remedy 9.8 - 2017	529999.00
	Road Bikes	Trek Domane SLR 9 Disc - 2018	1199999.00



Total **13** rows returned.

This shows the highest-priced product available in each category.



KEY INSIGHT: These are the premium products in each category, which can help in pricing strategy, inventory planning, and business decision-making.



QUESTION 6

TOTAL NUMBER OF ORDERS PLACED BY EACH CUSTOMER PER STORE



QUESTION

Find the total number of orders placed by each customer per store.



SQL QUERY

```
SELECT
    CONCAT(c.first_name, ' ', c.last_name) AS Customers_Name,
    s.store_name,
    COUNT(o.order_id) AS Total_Orders
FROM
    customers c
JOIN
    orders o USING (customer_id)
JOIN
    stores s USING (store_id)
GROUP BY Customers_Name , s.store_name
ORDER BY Total_Orders DESC;
```



Tables Used: customers, orders, stores



RESULT (Sample)

Customers_Name	store_name	Total_Orders
▶ Monika Berg	Santa Cruz Bikes	3
Araceli Golden	Santa Cruz Bikes	3
Williemae Holloway	Santa Cruz Bikes	3
Corene Wall	Santa Cruz Bikes	3
Charolette Rice	Santa Cruz Bikes	3
Jamaal Albert	Santa Cruz Bikes	3
Saturnina Garner	Santa Cruz Bikes	3
Deloris Burke	Santa Cruz Bikes	3
Ronna Butler	Santa Cruz Bikes	3
...	...	...



Total **1000+** rows returned.

This shows the total number of orders placed by each customer in each store.



KEY INSIGHT: This analysis helps in understanding customer engagement across different stores and identifying high-activity customers in each location.



QUESTION 7

NAMES OF STAFF MEMBERS WHO HAVE NOT HANDLED ANY ORDER



QUESTION

Find the names of staff members who have not handled any order.



SQL QUERY

```
SELECT
    CONCAT(s.first_name, ' ', s.last_name) AS staff_name,
    o.order_id
FROM
    staffs s
LEFT JOIN
    orders o USING (staff_id)
WHERE
    order_id IS NULL;
```



Tables Used: staffs, orders



RESULT

	staff_name	order_id
▶	Fabiola Jackson	NULL
	Virgie Wiggins	NULL
	Jannette David	NULL
	Bernardine Houston	NULL



Total **4** rows returned.
These staff members have not handled any order.



KEY INSIGHT: Helps identify underutilized staff members who are not currently assigned to any orders, enabling better resource allocation.



QUESTION 8

TOP 3 MOST SOLD PRODUCTS IN TERMS OF QUANTITY



QUESTION

Find the top 3 most sold products in terms of quantity.



SQL QUERY

```
SELECT
    p.product_name, SUM(oi.quantity) AS Total_Quantity_Sold
FROM
    products p
    JOIN
    order_items oi USING (product_id)
GROUP BY p.product_name
ORDER BY Total_Quantity_Sold DESC
LIMIT 3;
```



Tables Used: products, order_items



RESULT

	product_name	Total_Quantity_Sold
▶	Electra Cruiser 1 (24-Inch) - 2016	296
	Electra Townie Original 7D EQ - 2016	290
	Electra Townie Original 21D - 2016	289



Total **3** rows returned.
These are the top 3 most sold products by quantity.



KEY INSIGHT: The top 3 most sold products by quantity indicate the highest demand items, helping in inventory planning and ensuring product availability.



QUESTION 9

FIND THE MEDIAN VALUE OF THE PRICE LIST



QUESTION

Find the median value of the price list.



SQL QUERY

```
SELECT
    list_price AS median_price
FROM
    products
ORDER BY list_price
LIMIT 1 OFFSET 160;
```



Tables Used: products



RESULT

median_price	
▶	74999.00



Total **1** row returned.
This is the median value of the price list.



KEY INSIGHT: The median price helps in understanding the central value of the price list, reducing the impact of extremely high or low prices.



QUESTION 10

LIST ALL PRODUCTS THAT HAVE NEVER BEEN ORDERED (USE EXISTS)



QUESTION

List all products that have never been ordered.



SQL QUERY

```
SELECT
    p.product_id, p.product_name
FROM
    products p
WHERE
    NOT EXISTS( SELECT
        1
    FROM
        order_items oi
    WHERE
        p.product_id = oi.product_id);
```



Tables Used: products, order_items



RESULT (Sample)

	product_id	product_name
▶	1	Trek 820 - 2016
	121	Surly Krampus Frameset - 2018
	125	Trek Kids' Dual Sport - 2018
	154	Trek Domane SLR 6 Disc Women's - 2018
	195	Electra Townie Go! 8i Ladies' - 2018
	267	Trek Precaliber 12 Girl's - 2018
	284	Electra Savannah 1 (20-inch) - Girl's - 2018
	291	Electra Sweet Ride 1 (20-inch) - Girl's - 2018
	...	...
	321	Trek Checkpoint ALR Frameset - 2019



Total **14** rows returned.
These are products that have never been ordered.



KEY INSIGHT: Identifying products that have never been ordered helps in evaluating product performance, demand gaps, and making informed inventory decisions.



QUESTION 11

STAFF MEMBERS WHO HANDLED MORE ORDERS THAN THE AVERAGE



QUESTION

List the names of staff members who have handled more orders than the average number of orders handled by all staff members.



SQL QUERY

```
SELECT
    CONCAT(s.first_name, ' ', s.last_name) AS staff_name,
    COUNT(o.order_id) AS Total_Sales
FROM
    staffs s
JOIN
    orders o USING (staff_id)
GROUP BY s.staff_id, staff_name
HAVING COUNT(o.order_id) > (SELECT
    AVG(staff_total)
FROM
    (SELECT
        COUNT(o.order_id) AS Staff_Total
    FROM
        staffs s
    LEFT JOIN orders o USING (staff_id)
    GROUP BY staff_id) AS avg_sale);
```



Tables Used: staffs, orders



RESULT

	staff_name	Total_Sales
▶	Mireya Copeland	164
	Genna Serrano	184
	Marcelene Boyer	553
	Venita Daniel	540



Total **4** rows returned.
These staff members handled more orders than the average number of orders handled by all staff members.



KEY INSIGHT: These staff members consistently handled more orders than the overall staff average, highlighting top-performing employees who can be recognized for operational efficiency and workload management.



QUESTION 12

IDENTIFY THE CUSTOMERS WHO HAVE ORDERED ALL TYPES OF PRODUCTS FROM ROAD BIKES CATEGORY



QUESTION

Identify the customers who have ordered all types of products from Road Bikes category.



SQL QUERY

```
SELECT
  customer_id,
  CONCAT(c.first_name, ' ', c.last_name) AS Customers_Name
FROM
  customers c
  JOIN orders o USING (customer_id)
  JOIN order_items oi USING (order_id)
  JOIN products p USING (product_id)
  JOIN categories ca USING (category_id)
WHERE category_name = 'Road Bikes'
GROUP BY customer_id
HAVING COUNT(DISTINCT p.product_id) = (SELECT COUNT(*) FROM products p
  JOIN categories ca USING (category_id)
  WHERE category_name = 'Road Bikes');
```



Tables Used: customers, orders, order_items, products, categories



RESULT

customer_id	Customers_Name
 No result found.	



No customers have ordered all types of products from the Road Bikes category.



KEY INSIGHT: No customer has purchased all product types from the Road Bikes category. This indicates product variety purchase is still incomplete by any single customer.





KEY FINDINGS

Insights from SQL Analysis of Bicycle Sales Database



Baldwin Bikes is the **top performing store** with the highest sales and revenue.



Pamela Newman is the **highest spending customer**, showing strong loyalty and purchasing behavior.



Electra Cruiser 1, Electra Townie 7D EQ and Electra Townie 21D are the **top selling products** across all stores.



The **median product price** is ₹74,999 with highest priced products concentrated in 7 categories.



4 **staff performance** handled more orders than average, while 4 staff had no assigned orders.



14 products were **never ordered**, indicating slow-moving inventory and optimization opportunity.



Overall Insight:

The analysis uncovers key opportunities to **boost sales**, **improve customer retention**, **optimize inventory**, and **drive data-driven decisions** for business growth.





BUSINESS RECOMMENDATIONS



Strategic actions to improve sales performance, customer engagement and operational efficiency.



Focus on **top performing stores** to drive higher sales.



Optimize staff productivity and ensure better task allocation.



Retain high value customers with loyalty programs and offers.



Improve inventory management by reducing slow-moving stock.



Promote top selling products and maintain stock availability.



Expand in high demand markets to capture more opportunities.



Review pricing strategy to maintain competitiveness.



Use data-driven insights for better decision making.



Goal: Increase sales, improve customer retention, optimize inventory, and drive sustainable business growth.





PROJECT CONCLUSION

Key Takeaways from SQL Analysis of Bicycle Sales Database



DATA
DRIVEN GROWTH



Extracted meaningful insights on sales, customers, products, stores and staff performance using advanced SQL queries.



Identified key revenue drivers and opportunities for business growth.



Detected inventory gaps and slow-moving items to optimize stock management.



Analyzed pricing patterns to support better pricing and product strategy.



Evaluated staff productivity and order handling to improve operational efficiency.



Built a strong foundation for data-driven decision making and future business planning.



This project showcases the power of SQL in turning raw data into **actionable insights** that drive **better decisions** and support overall **business success**.





THANK YOU!



Thank you for your time and attention.
Hope you found the insights from SQL Analysis of
Bicycle Sales Database
valuable and insightful.



**Data-Driven
Insights**



**Customer & Sales
Understanding**



**Inventory
Optimization**



**Pricing
Intelligence**



**Better Decision
Making**



Turning data into **actionable insights**
to drive better business outcomes.